Ex1: It is asked for estimation of the lamps' defection proportion. In order that the difference of the exact and estimated proportion is not more than 0.03, find the size of the sample for %95 of confidence interval

a) if we know that the sample proportion is 0.05

a)

$e = 0.03$

$\alpha = 0.5$

$Z_{\frac{0.5}{2}} = 1.96$

$$z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}\hat{q}}{n}}}$$

a) $\hat{p} = 0.05$

$\hat{q} = 0.95$

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{pq}{n}} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\frac{pq}{n}}$$

$$e = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} = 1.96\sqrt{\frac{0.05 \times 0.95}{n}} = 0.03$$

$$\Rightarrow n = \left[\frac{1.96^2}{1} \cdot \frac{0.05 \times 0.95}{0.03^2}\right] = 202.751$$

We need a sample of size 203.

b) If we have no idea about the sample proportion, then (don't) we will select a value of $\hat{p}$ so as to maximize n. Thus,

for $n = \left[z_{\alpha/2}^2 \frac{\hat{p}_1 \hat{q}_1}{e^2}\right]$ $\hat{p}_1 = \hat{q}_1 = \frac{1}{2}$ maximize it.

$$\Rightarrow n = z_{\alpha/2}^2 \frac{1}{4e^2} = 1.96^2 \frac{1}{4 \cdot 0.03^2} = 1067.111$$

We need a sample of size 1068.

Ex2: The diameter of the screws prodiced in a factory has normal distribution with the standard deviation 0.38 cm. In order to estimate the mean of screw diameters, 12 random screws are selected and 95% of confidence interval is computed as

$$1.64 \leq \mu \leq 2.36.$$

However, the factory manager asks more definite interval. In order that the difference of the exact value and estimated value is not more than 0.08 cm, find the size of the sample for

a) for 95% confidence interval,

b) // 99% // //.

a) $\sigma = 0.38$

$n = 12$ $\quad Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

$\alpha = 0.05 \Rightarrow z_{\alpha/2} = 1.96$

$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \sigma/\sqrt{n}$$

$$e = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 0.08$$

$$\Rightarrow 1.96 \frac{0.38}{\sqrt{n}} = 0.08 \Rightarrow n = \left(\frac{1.96 \times 0.38}{0.08}\right)^2 = 86.676$$

$$\Rightarrow n = 87$$

We need a sample of size 87.

b) $\alpha = 0.01 \Rightarrow z_{0.005} \cong 2.58$

$$n = \left[\frac{2.58 \cdot 0.38}{0.08}\right]^2 = 150.185$$

$$\Rightarrow n = 151$$

Ex 3: $X \sim N(8, 2^2)$, $n = 25$

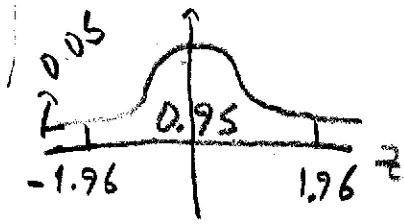
$$P(7.8 < X < 8.2) = P\left(\frac{7.8-8}{2/\sqrt{25}} < z < \frac{8.2-8}{2/\sqrt{25}}\right) = P(-0.5 < z < 0.5)$$
$$= 2 \times 0.1915 = 0.3830$$
$$= 38.3\%$$

Ex 4:
$X \sim N(\mu, 8^2)$

$$P(|\bar{X} - \mu| \leqslant 1.5) = 0.95$$

$$\Rightarrow P\left(|z| \leqslant \frac{1.5}{\sigma/\sqrt{n}}\right) = 0.95$$

$$\Rightarrow P\left(-\frac{1.5}{8}\sqrt{n} \leqslant z \leqslant \frac{1.5}{8}\sqrt{n}\right) = 0.95$$



$$\Rightarrow 1.96 = \frac{1.5}{8}\sqrt{n} \Rightarrow \sqrt{n} = \left[\frac{1.96 \times 8}{1.5}\right] \Rightarrow n = \left[\underbrace{\frac{1.96 \times 8}{1.5}}_{109.27}\right] \cong 110$$

Ex 5:
$n = 400$ (without repl.)
$N = 5000$
$\sigma = 50$
$\bar{x} = 4500$
$\alpha = 0.10$

$$\frac{n}{N} = \frac{400}{5000} = 0.08 > 0.05$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}} = \frac{50}{\sqrt{400}}\sqrt{\frac{5000-400}{5000-1}} \cong 2.3982$$

$n > 30$

$$\left[\bar{X} \mp \underbrace{z_{\alpha/2}}_{1.645} \frac{\sigma}{n}\sqrt{\frac{N-n}{N-1}}\right] = [4500 \mp 1.645 \times 2.3982]$$
$$= [4496.1, 4503.9]$$

Ex 6:
$N = 10000$
$n = 36$ (without rep.) $\quad \frac{n}{N} = \frac{36}{10000} = 0.0036 < 0.05 \Rightarrow \sigma_{\bar{x}} = \frac{s}{\sqrt{n}}$
$\bar{x} = 20$ gr
$s = 4$ gr
$\alpha = 0.30$

Since $n > 30$,

$$\left[\bar{X} \mp \underbrace{z_{0.15}}_{1.04} \frac{s}{\sqrt{n}}\right] = \left[20 \mp 1.04 \frac{4}{\sqrt{36}}\right]$$
$$= [19.307, 20.693]$$

Ex 7: $N = 5000$, $n = 49$

$\sum_{i=1}^{49} X_i = 2940$

$\alpha = 0.05$

$\sigma = 7$

$\frac{n}{N} = \frac{49}{1000} < 0.05$

$\Rightarrow \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

$n > 30$

$\left[\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$

$\bar{X} = \frac{\sum x_i}{n} = \frac{2940}{49} = 60$

$\left[60 \mp 1.96 \frac{7}{\sqrt{49}} \right]$

$[58.04, 61.96]$

Ex 8

$n = 180$ without rep

$N = 2000$

$\frac{n}{N} = \frac{180}{2000} = 0.09 > 0.05$

$X = 27$

$\alpha = 0.01$

$\sigma_{\hat{p}} = \sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}}$

$n > 30$

$\left[\hat{p} \mp z_{0.005} \underbrace{\sqrt{\frac{0.15 \times 0.85}{180}} \sqrt{\frac{2000-180}{2000-1}}} \right]$

$\hat{p} = \frac{27}{180} = 0.15$

$\hat{q} = 1 - 0.15 = 0.85$

$= [0.15 \mp 2.575 \times 0.0254]$

$= [0.0846, 0.2154]$

Ex 9:

$X \sim N$, $n = 12$, $S = 16.8$, $\alpha = 0.10$

$\chi^2_{0.05,11} = 19.68$ $\quad \chi^2_{0.95,11} = 4.57$

$\left[\frac{(12-1)16.8^2}{19.68}, \frac{(12-1)16.8^2}{4.57} \right] = [157.8, 679.4]$

Ex 10:

$\hat{p} = \frac{72}{225} = 0.32$

$\left[\hat{p} \mp z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} \right]$

$n = 225$

$\left[0.32 \mp 2.575 \sqrt{\frac{0.32 \times 0.68}{225}} \right] \cong [0.24, 0.40]$

$\alpha = 0.01 \Rightarrow z_{0.005} = 2.575$

$\hat{q} = 1 - \hat{p} = 0.68$

Ex.11: $n_1 = 55$, $\bar{X}_1 = 20000$, $S_1 = 950$ $\quad X_1 \sim N$ $\quad \alpha = 0.05$ (5)

$n_2 = 48$, $X_2 = 17000$, $S_2 = 1000$ $\quad X_2 \sim N$

$$\left[(\bar{X}_1 - \bar{X}_2) \mp t_{\vartheta} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right] \qquad \vartheta = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}} \approx 98$$

$$\Downarrow$$

$$\left[(20000 - 17000) \mp 1.96 \sqrt{\frac{950^2}{55} + \frac{1000^2}{48}} \right] \quad t_{98, 0.025} \approx z_{0.025} = 1.96$$

$$\Downarrow$$

$[2621.75, 3378.25]$ $\Rightarrow$ significant difference

Ex.12

$\hat{p}_1 = \frac{60}{75}$, $\hat{q}_1 = \frac{15}{75}$, $\alpha = 0.01$

$\hat{p}_2 = \frac{55}{80}$, $\hat{q}_2 = \frac{25}{80}$ $\quad \left[(\hat{p}_1 - \hat{p}_2) \mp z_{0.005} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \right]$

$\Rightarrow [-0.07, 0.29] \Rightarrow$ no significant difference

Ex. 13 $X \sim N$, $n = 8$, $\alpha = 0.05$, $z_{0.025} = 1.96$

$\bar{X} = \frac{\sum X_i}{8} = 10.375$

a) $\left[\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] \Rightarrow \left[10.375 \mp 1.96 \frac{\sqrt{3}}{\sqrt{8}} \right] \Rightarrow [9.175, 11.575]$

b) $\left[\bar{X} \mp t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \right] \Rightarrow \left[10.375 \mp 2.365 \frac{\sqrt{3.411}}{\sqrt{8}} \right] \Rightarrow [8.881, 11.919]$

$S = \sqrt{\frac{\sum (X_i - \bar{X})^2}{8 - 1}} = \sqrt{3.411}$

c) $\left[\bar{X} \mp t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \right] = \left[10.375 \mp 2.083 \sqrt{\frac{3.411}{8}} \right] = [9.015, 11.735]$

$\alpha = 0.04$

$$\left.\begin{array}{l} 2.365 \rightarrow 0.025 \\ t \rightarrow 0.04 \\ 1.895 \rightarrow 0.05 \end{array}\right\} \quad \frac{2.365 - 1.895}{0.025 - 0.05} = \frac{2.365 - t}{0.025 - 0.04} \Rightarrow t = 2.083$$

Ex 14 $n = 35$, $\sigma = 28$

a) $P[|\bar{X} - \mu| \leq 8] = P\left[|z| \leq \frac{8}{\sigma/\sqrt{n}} \right] = P\left[|z| < \frac{8}{28/\sqrt{35}} \right]$

$= P(-1.69 < z < 1.69) = 2 \times 0.4545 = 0.9090$

b) 

$z_{\alpha/2} = 2.575$ $\quad \frac{8}{28/\sqrt{n}} = 2.575 \Rightarrow n = \lceil 81.225 \rceil \Rightarrow n = 82$

Ex 15: $p=0.25 \quad , n=40$

$q=1-p=0.75$

$$P[\hat{p}>0.27] = P\left[\frac{\hat{p}-p}{\sqrt{pq/n}} > \frac{0.27-p}{\sqrt{pq/n}}\right] = P\left(z > \frac{0.27-0.25}{\sqrt{\frac{0.25\times0.75}{40}}}\right)$$

$$= P(z>0.29) = 0.5-0.1141 = 0.3859$$

Ex 16

$X \sim N \quad , n=16 \quad \to$ claim $\sigma^2=32$

If $s^2<13.355$ or $s^2>65.280 \to$ the claim is rejected

$$P(s^2<13.355) = P\left(\frac{(n-1)s^2}{\sigma^2} < \frac{13.355\,(16-1)}{32}\right) = P(\chi^2_{15}<6.26)$$

$$= 0.025$$

$$P(s^2>65.280) = P\left(\frac{(n-1)s^2}{\sigma^2} > \frac{65.280\,(16-1)}{32}\right) = P(\chi^2_{15}>30.6)$$

$$= 0.010$$

$$\Rightarrow P(s^2<13.355 \text{ or } s^2>65.280) = 0.025+0.010 = 0.035$$

Ex. 17 $n=100, \ \bar{X}=1040, \ s=25, \ \alpha=0.05 \ \Rightarrow z_{0.025}=1.96$

$n\geq 30$ $\quad \left[\bar{X} \mp z_{\alpha/2}\frac{s}{\sqrt{n}}\right] = \left[1040 \mp 1.96\frac{25}{\sqrt{100}}\right] = [1035.1, 1044.9]$

Ex. 18 $n=25, \ \bar{X}=50$

$X\sim N, \ \sigma=10 \quad \alpha=0.05 \quad , z_{0.025}=1.96$

$$\left[\bar{X} \mp z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] = \left[50 \mp 1.96\frac{10}{\sqrt{25}}\right] = [46.08, 53.92]$$

Ex 19: $n=100$

$\bar{X}=1280 \quad \alpha=0.05$

$s=140 \quad n\geq 30 \Rightarrow \left[\bar{X} \mp z_{\alpha/2}\frac{s}{\sqrt{n}}\right] = \left[1280 \mp 1.96\frac{140}{\sqrt{100}}\right]$

$= [1252.56, 1307.44]$

Ex 20: $n=400 \quad , \alpha=0.05 \quad , z_{0.025}=1.96$

$\hat{p}=\frac{32}{400}=0.08$

$\hat{q}=1-\hat{p}=0.92$ $\quad \left[\hat{p} \mp z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}\right] = \left[0.08 \mp 1.96\sqrt{\frac{0.08\times0.92}{400}}\right]$

$\simeq [0.053, 0.107]$

Ex.21 $n_1=40$ $n_2=35$ $\alpha=0.01$

$\bar{x}_1=86$ $\bar{x}_2=72$ $n_1\geq 30,\ n_2\geq 30$

$s_1=12$ $s_2=14$

$z_{0.005}=2.575$

$$\left[(\bar{x}_1-\bar{x}_2)\mp z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}\right]=\left[(86-72)\mp 2.575\sqrt{\frac{12^2}{40}+\frac{14^2}{35}}\right]$$

$$\cong[6.19,\ 21.81]$$

Ex.22 $n_1=n_2=1000$

$\hat{p}_1=\frac{825}{1000}=0.825 \Rightarrow \hat{q}_1=0.175$

$\hat{p}_2=\frac{760}{1000}=0.76 \Rightarrow \hat{q}_2=0.24$ $\alpha=0.05 \Rightarrow z_{0.025}=1.96$

$$(\hat{p}_1-\hat{p}_2)-z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}\leq p_1-p_2\leq(\hat{p}_1-\hat{p}_2)+z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}$$

$$\Rightarrow\left[(0.825-0.76)\mp 1.96\sqrt{\frac{0.825\times0.175}{1000}+\frac{0.76\times0.24}{1000}}\right]$$

$$=[0.0296,\ 0.1004]$$

Ex.23 $n=16$ $\bar{x}=35$

$X\sim N$ $s=4.6$ $\alpha=0.05$, $t_{0.025,16-1}=2.13145$

$$\left[\bar{x}\mp t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}\right]=\left[35\mp 2.13145\,\frac{4.6}{\sqrt{16}}\right]\Rightarrow 32.55\leq\mu\leq 37.45$$

Ex.24 $\bar{x}=20500$

$s=1600$

$n=8$

$\alpha=0.10$

$$\Rightarrow\left[\frac{(n-1)s^2}{\chi^2_{n-1,0.05}},\ \frac{(n-1)s^2}{\chi^2_{n-1,0.95}}\right]=\left[\frac{7\times1600^2}{14.06714},\ \frac{7\times1600^2}{2.16735}\right]$$

$$\cong[1273890.8,\ 8268161.8]$$

$$\Downarrow$$

$$1128.67\leq\sigma\leq 2875.44$$

$\chi^2_{7,0.05}=14.06714$

$\chi^2_{7,0.95}=2.16735$

Ex.25: $n_1=10$ $n_2=11$

$\bar{x}_1=1.5$ $\bar{x}_2=2.0$

$\alpha=0.02$ $s_1=0.4$ $s_2=0.6$

$F_{0.01,9,10}=4.94$

$F_{0.01,10,9}=5.26$

$$\frac{1}{F_{\alpha/2,9,10}}\frac{s_1^2}{s_2^2}\leq\frac{\sigma_1^2}{\sigma_2^2}\leq F_{\alpha/2,10,9}\frac{s_1^2}{s_2^2}$$

$$\Rightarrow\frac{1}{4.94}\frac{0.4^2}{0.6^2}\leq\frac{\sigma_1^2}{\sigma_2^2}\leq 5.26\,\frac{0.4^2}{0.6^2}$$

$$\Rightarrow 0.09\leq\sigma_1^2/\sigma_2^2\leq 2.34$$

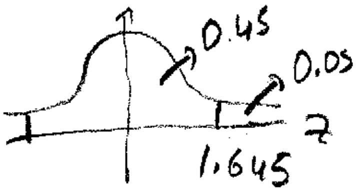
$$\Rightarrow 0.30\leq\frac{\sigma_1}{\sigma_2}\leq 1.53$$

Ex.26: $\hat{p} = 0.40$

$n = ?$

$$p(|\hat{p}-p| = 0.04) = 0.90$$

$$\Rightarrow p\left(\frac{|\hat{p}-p|}{\sqrt{\frac{pq}{n}}} = \frac{0.04}{\sqrt{pq/n}}\right) = 0.90$$



$$\Rightarrow P\left(|z| = \frac{0.04}{\sqrt{pq/n}}\right) = 0.90 \Rightarrow P\left(-\frac{0.04}{\sqrt{pq/n}} \leq z \leq \frac{0.04}{\sqrt{\frac{0.4\times 0.6}{n}}}\right) = 0.90$$

$$\Rightarrow 1.645 = \frac{0.04}{\sqrt{\frac{0.4\times 0.6}{n}}} \Rightarrow n = \left\lceil \frac{1.645^2}{0.04^2}\, 0.4\times 0.6 \right\rceil \simeq \lceil 405.9 \rceil = 406$$

Ex 27:

$N = 500$ (without replacement)

$n = 49$

$n \geq 30$

$z_{\alpha/2} = z_{0.025} = 1.96$

$$\frac{n}{N} = \frac{49}{500} = 0.098 > 0.05 \Rightarrow \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

$$\left[\bar{x} \mp z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}\right] = \left[60 \mp 1.96\,\frac{7}{\sqrt{49}}\sqrt{\frac{500-49}{500-1}}\right]$$

$$\bar{x} = \frac{\sum x_i}{49} = \frac{2940}{49} = 60$$

$$= [58.14, 61.86]$$

Ex28:

$n = 64$

$N = 6000$

$$\frac{n}{N} = \frac{64}{6000} = 0.0107 \leq 0.05$$

$$\bar{x} = \frac{\sum x_i}{64} = \frac{3200}{64} = 50$$

$$s^2 = \frac{\sum_{i=1}^{64}(x_i-\bar{x})^2}{64-1} = \frac{1575}{63} = 25$$

$$\Rightarrow s = 5$$

$\alpha = 0.01$, $n \geq 30$, $z_{0.005} = 2.575$

$$\left[\bar{x} \mp z_{\alpha/2}\frac{s}{\sqrt{n}}\right] = \left[50 \mp 2.575\frac{5}{8}\right]$$

$$\simeq [48.4, 51.6]$$

Ex 29:

$n_1 = 1000$ $\qquad n_2 = 1000$ $\qquad n_1 \geq 30,\ n_2 \geq 30$

$\hat{p}_1 = \frac{52}{1000} = 0.052$ $\qquad \hat{p}_2 = \frac{23}{1000} = 0.023$ $\qquad \alpha = 0.01$

$\hat{q}_1 = 1-\hat{p}_1 = 0.948$ $\qquad \hat{q}_2 = 1-\hat{p}_2 = 0.977$ $\qquad z_{\alpha/2} = z_{0.005} = 2.575$

$$\left[(\hat{p}_1-\hat{p}_2) \mp z_{0.005}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}\right] \simeq [0.007, 0.051]$$